



PIYUSH MAJI
Master of Science in Physics
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EDUCATION

Indian Institute of Technology Hyderabad
M.Sc. in Physics CGPA: 7.5/10

Expected July 2027

Hooghly Mohsin College, West Bengal
B.Sc. in Physics CGPA: 7.28/10

August 2024

RESEARCH

GNN-Based Full Event Interpretation — Belle II

Ongoing

Prof. Sourav Sandilya, Department of Physics, IIT Hyderabad

- Studying ML-driven event reconstruction for the Belle II experiment at KEK, focusing on graph neural networks (GNNs) for decay topology reconstruction inspired by the Full Event Interpretation (FEI) algorithm.
- Building detector-level graph representations of particle decay chains using Python and PyTorch.

PROJECTS

Diffraction Pattern Analysis for Wavelength Estimation

Completed

Prof. Bhuvanesh Ramakrishna, IIT Hyderabad [GitHub](#)

- Built a two-script Python pipeline to extract laser wavelength from camera images of diffraction gratings; preprocessed raw frames into 1D intensity profiles and fit the Fraunhofer grating model via non-linear least squares, obtaining $\lambda = 524 \pm 6$ nm.

COVID-19 Epidemic Modeling — SIR / SEIR / SUTRA

Completed

Prof. Anupam Gupta, IIT Hyderabad [GitHub](#)

- Implemented and compared SIR, SEIR, and SUTRA compartmental models on US COVID-19 data (Feb 2020 – Mar 2021); used NNLS regression for per-phase parameter estimation of contact rate $\tilde{\beta}$ and reach $\tilde{\rho}$, achieving MAPE as low as $\sim 4\%$ on individual phases. Ensembled all three models via linear regression.

Computational Modeling of Particle Motion in Brownian Systems

Completed

Prof. Anupam Gupta, IIT Hyderabad

- Simulated stochastic particle trajectories, computed Mean Squared Displacement (MSD), and performed log-log regression to identify diffusive regimes; validated results against theoretical predictions from statistical mechanics.

SKILLS

Languages & Tools: Python, NumPy, Pandas, SciPy, Matplotlib, Scikit-Learn, PyTorch, Git

Domain: Machine Learning, Graph Neural Networks, ODE/PDE Modeling, Statistical Inference, Scientific Computing, Data Analysis

Soft Skills: Research Communication, Critical Thinking, Scientific Writing, Collaborative Research

LEADERSHIP & EXTRACURRICULAR

Placement Coordinator

May 2026 – Present

Office of Career Services, IIT Hyderabad

- Serving as Placement Coordinator at OCS, IITH; managing placement operations, liaising with recruiting companies, and overseeing end-to-end coordination of campus recruitment drives.

Coordinator (Inbound and Multimedia)

Aug 2025 – Present

International Relations Cell, IIT Hyderabad

- Coordinated orientations, cultural programs, multimedia outreach, and outbound academic tours for international exchange students from Japan and the United States.